

Recursion

What is recursion ?

- function that calls itself until the problem is solved
- A base case is the point in which the problem is solved at

3 keys when function was called ?

1. Return address (where the function was called from),
2. Return value (the value to be returned), and
3. Arguments (the input parameters)

Step of recursion

1. Base case
2. Recurse
 1. Pre-operation (something done before recursion),
 2. Recursion (actual function call), and
 3. Post-operation (something done after recursion)

Function call stack ?

The stack grows downward as recursive calls are made, then unwinds upward as each function returns its calculated value, with each function adding its own computation to the final result

When use recursion over a for loop ?

When there is no defined end point, or when there is a significant branching factor that makes linear iteration difficult